

Data Analysis Project Report

Project Title: Flight Data Analysis – Understanding Factors Affecting Airline Ticket Prices

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Introduction:

The aviation industry generates vast amounts of data every day, making it an ideal domain for analytical exploration. This project aims to analyze a flight dataset to uncover meaningful insights related to ticket pricing and airline operations. By examining factors such as routes, departure and arrival times, seat classes, and booking timelines, this analysis provides a deeper understanding of how these variables influence flight prices. The insights derived from this study can be valuable for airline companies helping them optimize pricing strategies and travelers, who can use this information to make more informed booking decisions. Overall, this project demonstrates the use of data cleaning, exploration, and visualization techniques to extract actionable insights from real-world airline data.

Objective of this analysis:

The objective of this analysis is to get meaning full insight about the ticket prices of different airlines based on different factors such as source city, destination city, seat class, days left before the flights. We have also analyzed the pricing difference based on the arrival and departure time of the flights.

Basic Understanding of the Dataset:

The records in dataset are about multiple different airline operating in India. This data set shows the pricing and duration of each flight from different location based in India.

Number of columns available in dataset are 11.

Number of records available in dataset is 300153

Tools used for data Analysis:

1. Pandas: For data cleaning and analyzing
2. Matplotlib/Seaborn: For generating visualization of the analysis

Data Cleaning:

There was no need for cleaning the data as there were no missing and duplicate values.

Exploratory Data Analysis:

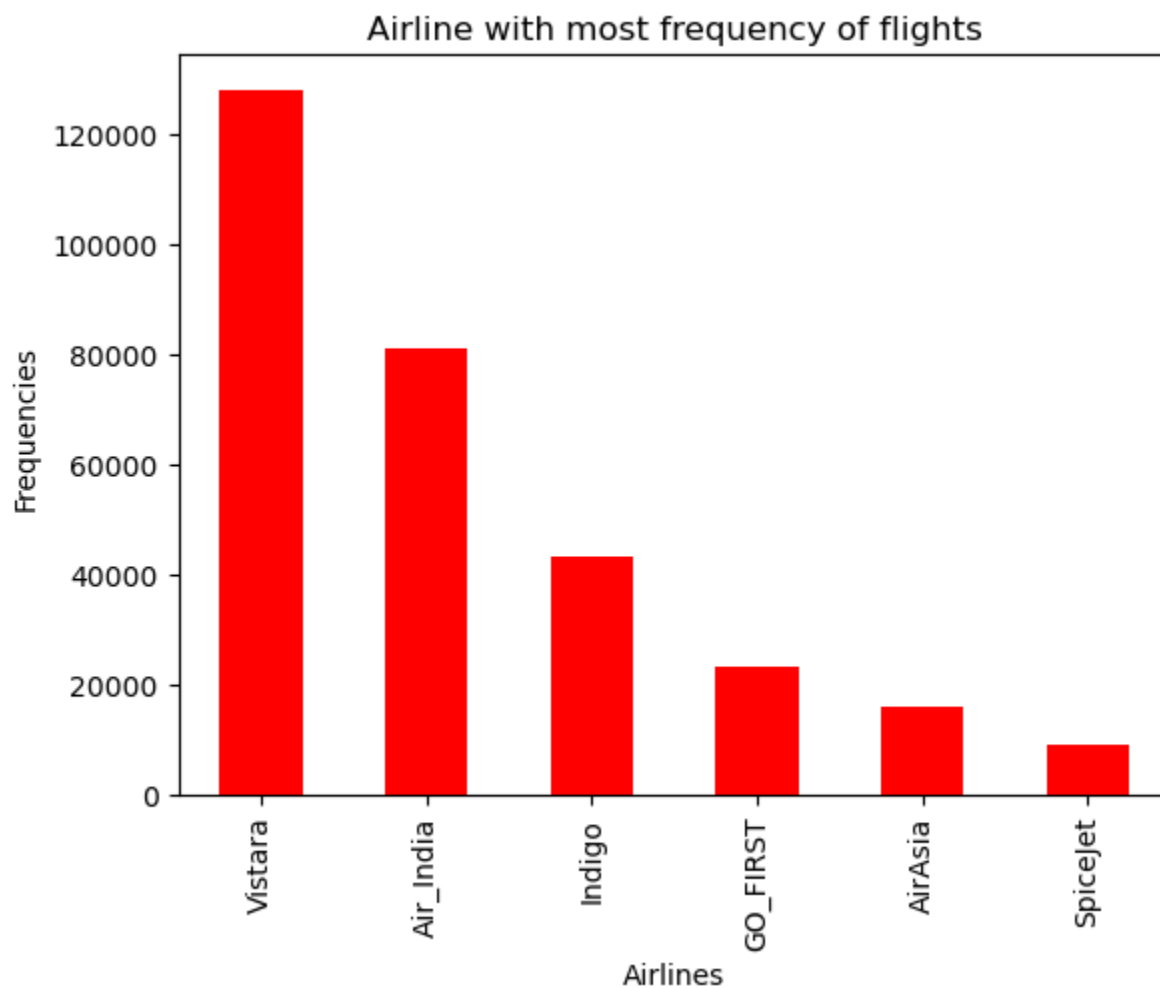
In this section we have solved different question to understand the behavior of ticket pricing based on different factor such as departure time, arrival time, seat class, source city, destination city etc.

We have solved 9 question and have generated some visualization for the better understanding of the data.

Q1. What are the airlines in the dataset, Accompanied by their frequency?

In this question we have tried to figure out the airline which has the most number of flights.

Bar graph:

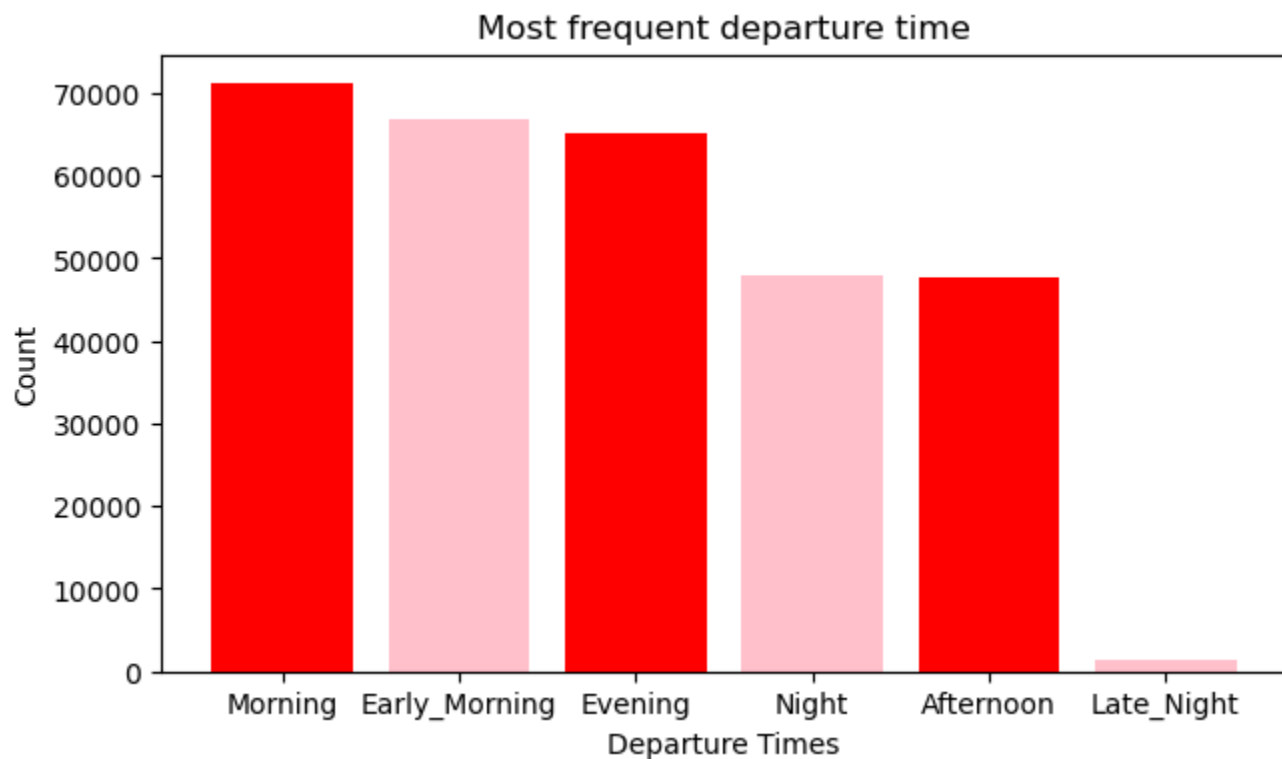


In this bar graph we can clearly see that Vistara airline has the most amount of flights.

Q2. Show bar graph representing the departure time and arrival time

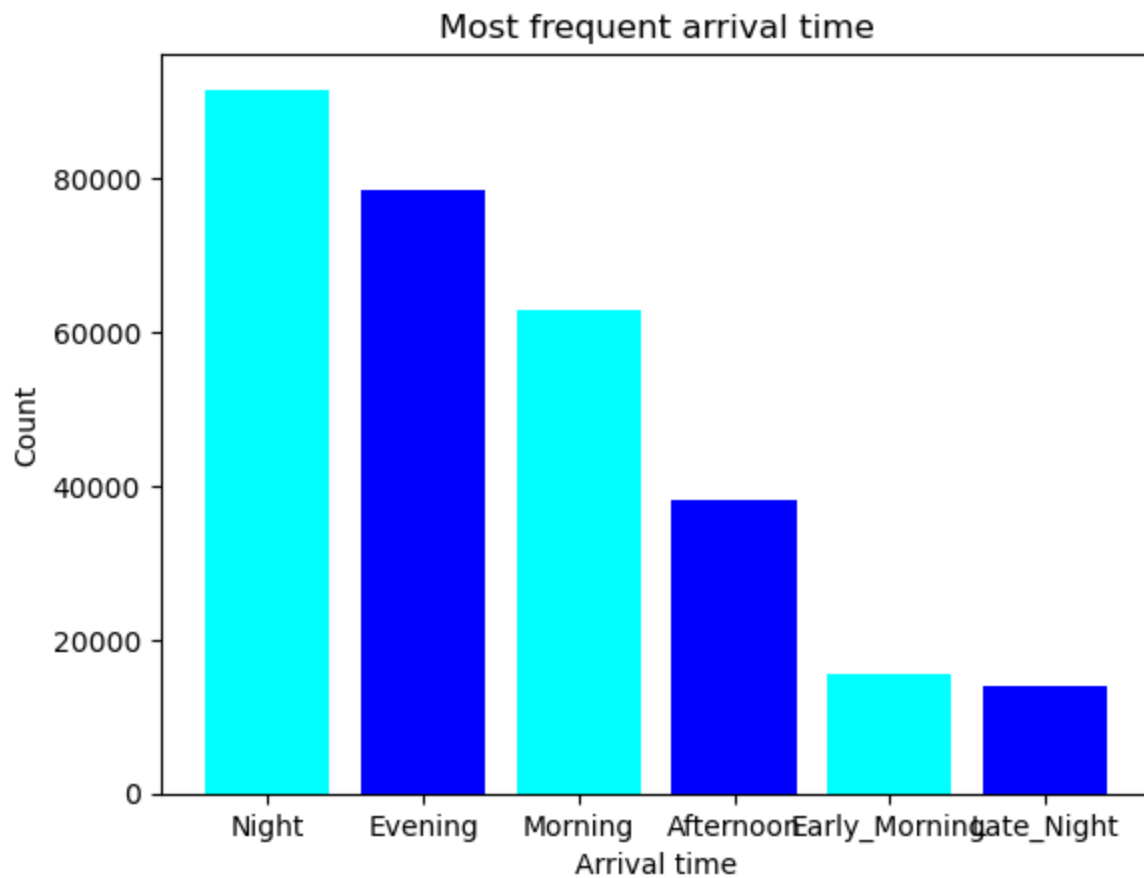
In this we have analyzed the most frequent departure and arrival time of all the airline by creating a bar graph

Bar Graph for departure time:



This bar graph shows the most frequent departure time. In this we can see that **Morning** is the most frequent departure time.

Bar Graph for Arrival time:

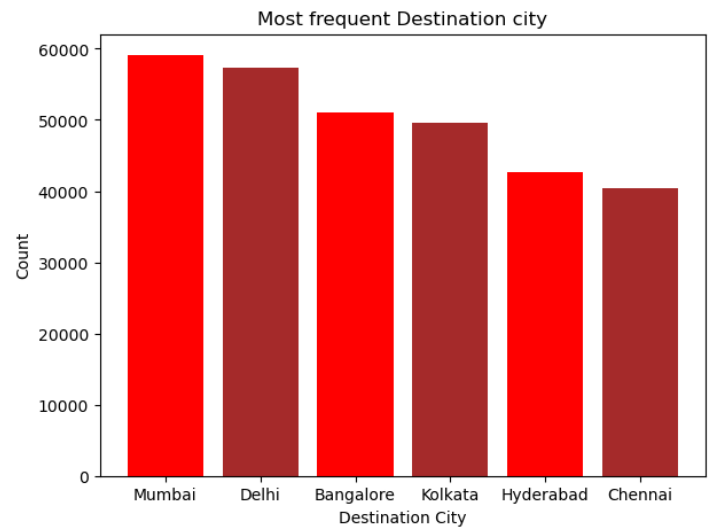
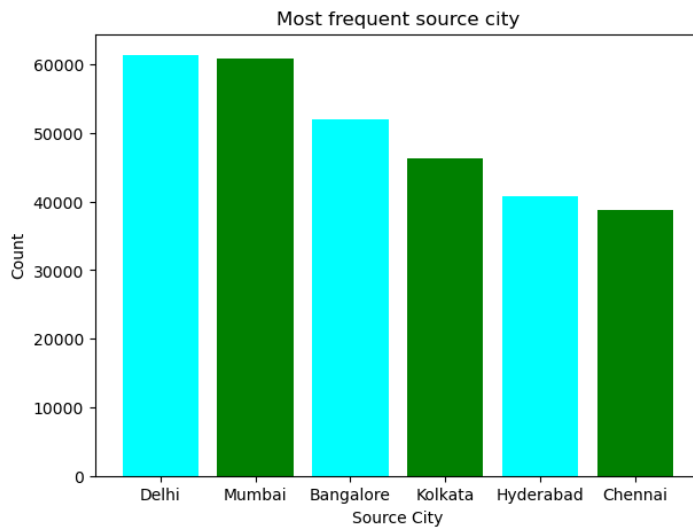


This bar graph shows the most frequent arrival time. In this graph we can see that **Night** is the most frequent arrival time.

Q3. Show bar graph representing the source city and destination city

In this we have analyzed the most frequent source and destination city preferred by the customer. We have created a bar graph for better understanding.

Bar graph for source city and destination city:



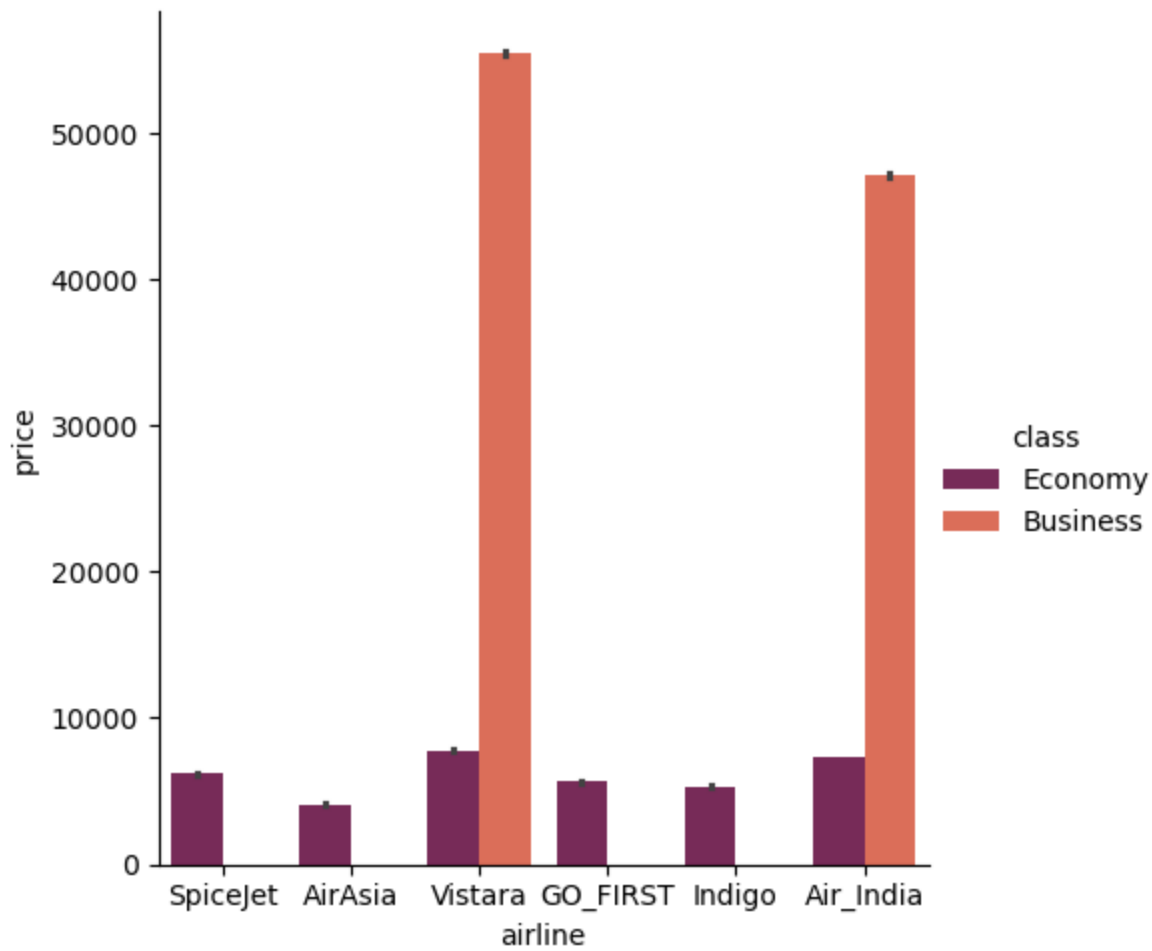
Above we have 2 bar graphs. The bar graph on the left shows the most frequent source city which is **Delhi** and the bar graph on the right shows the most frequent destination city which is **Mumbai**.

Q4. Does price vary with airlines?

In this we have tried to analyze the pricing of each airline to see if prices vary across different airlines based on the seat class whether it is economy or business class.

We have created a categorical plot for better understand of the price.

Chart:

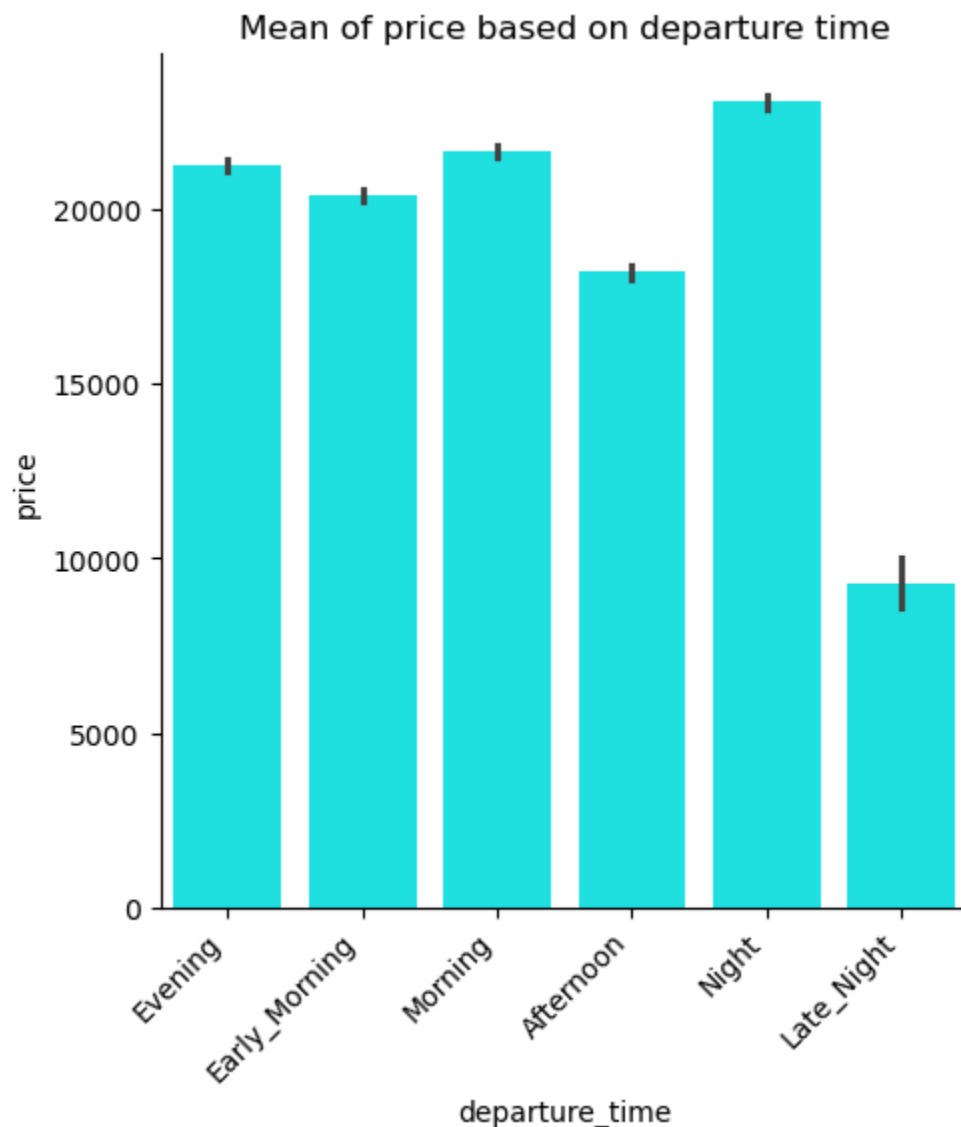


This is a categorical plot which shows the mean pricing for all the airline based on their seat class which is economy or business. In the above graph we can see that **Vistara** has the highest average price for both the seat class. Whereas **Air India** has the second highest average price for Business class.

Q5. Does tickets price change based on the arrival and departure time?

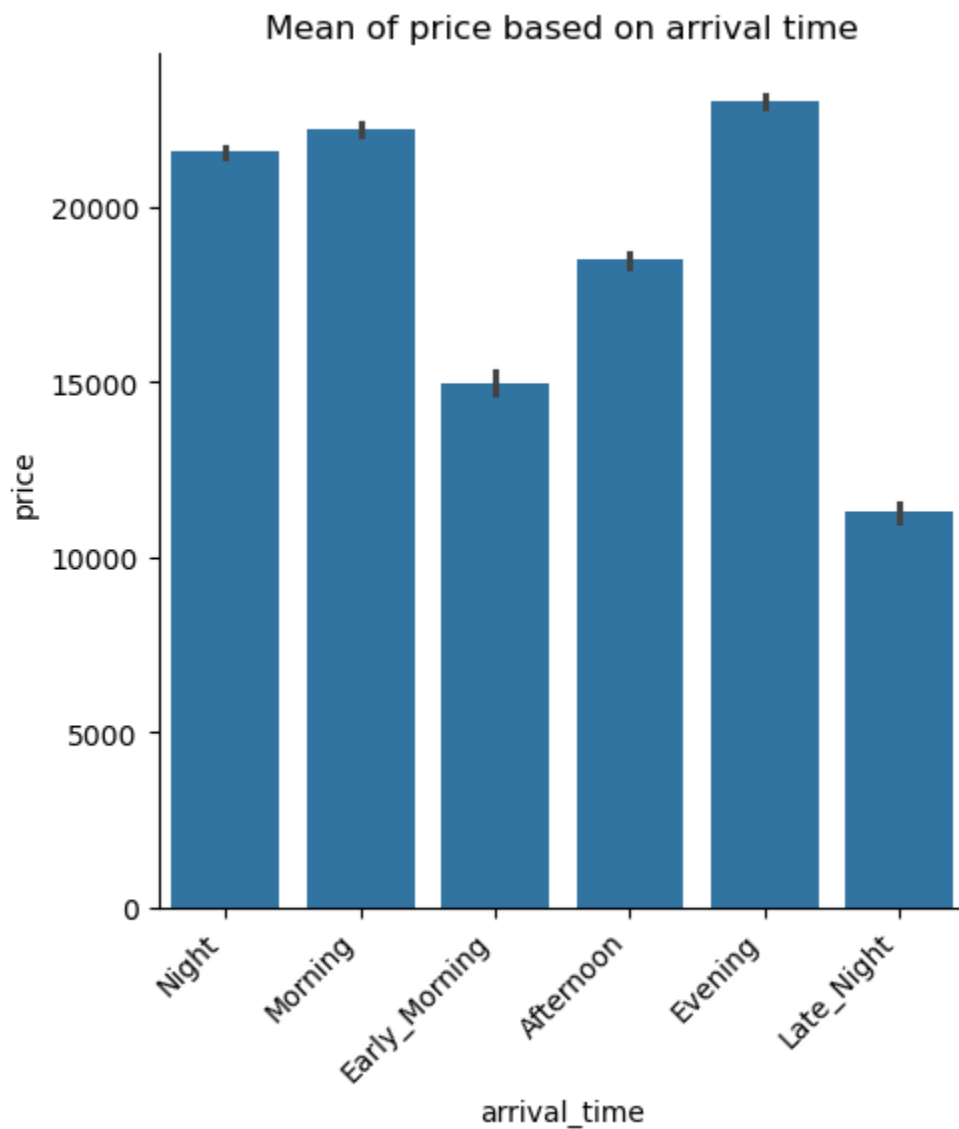
In this we have analyzed the pricing of the airline based on their arrival and departure timing. We have created a bar graph for both the timings

Bar graph for departure:



This graph shows the mean pricing for the departure time. In this we can see the pricing are usually higher for airlines when the departure time is **Night**.

Bar graph for arrival time:

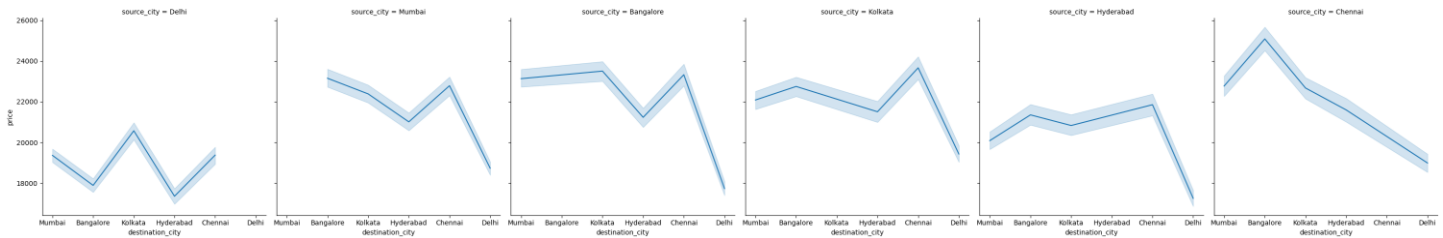


This graph shows the average price based on the arrival time of the airlines. In this we can see that **Evening** arrival time has the highest ticket price.

Q6. How the price changes with change in source and destination?

In this we have analyzed the ticket pricing based on the source and destination city. We have generated a relation plot for each source city comparing it with different destinations

Relation Plot:

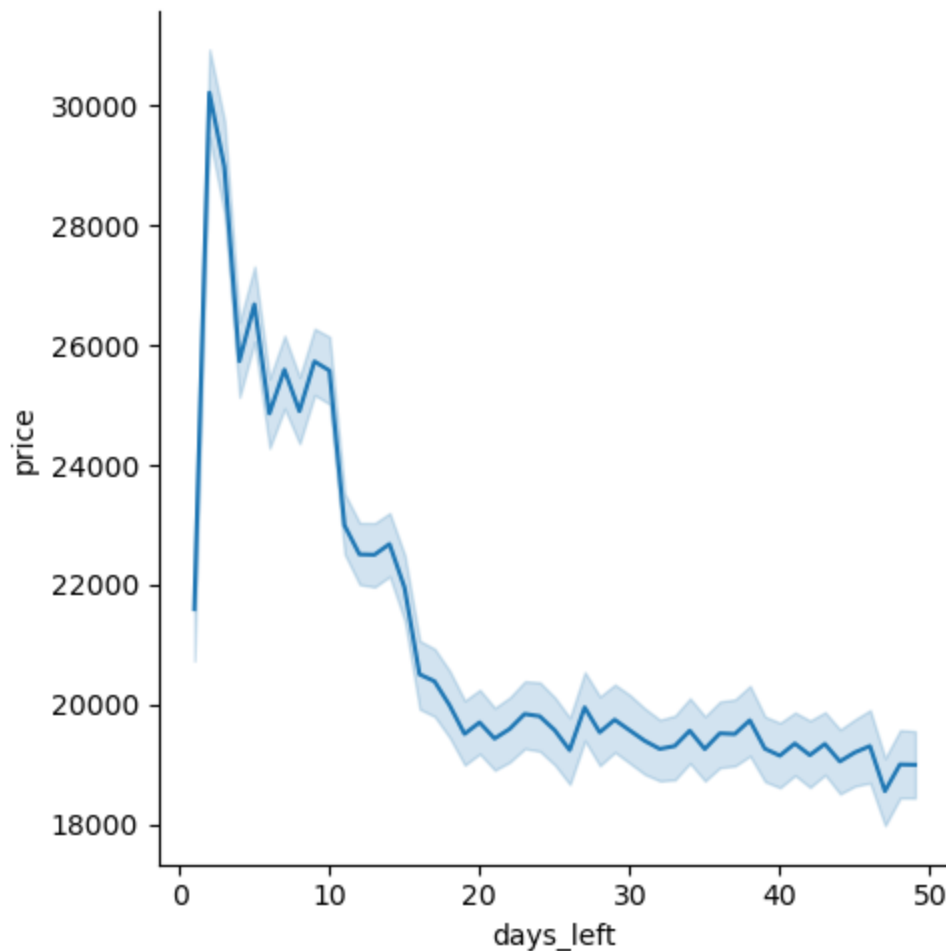


This graph shows the relation of each and every source city with different destination city. From this graph we have concluded that whatever the source city will be the price for Delhi as a destination will always be lower compared to other cities.

Q7. How are tickets price affected when tickets are brought in just 1 or 2 days before departure?

In this we have analyzed the pricing of tickets based the purchase before the flight. This analysis shows the relation between the pricing of tickets and days left for the flight.

We have created a line chart for better understanding of the data.

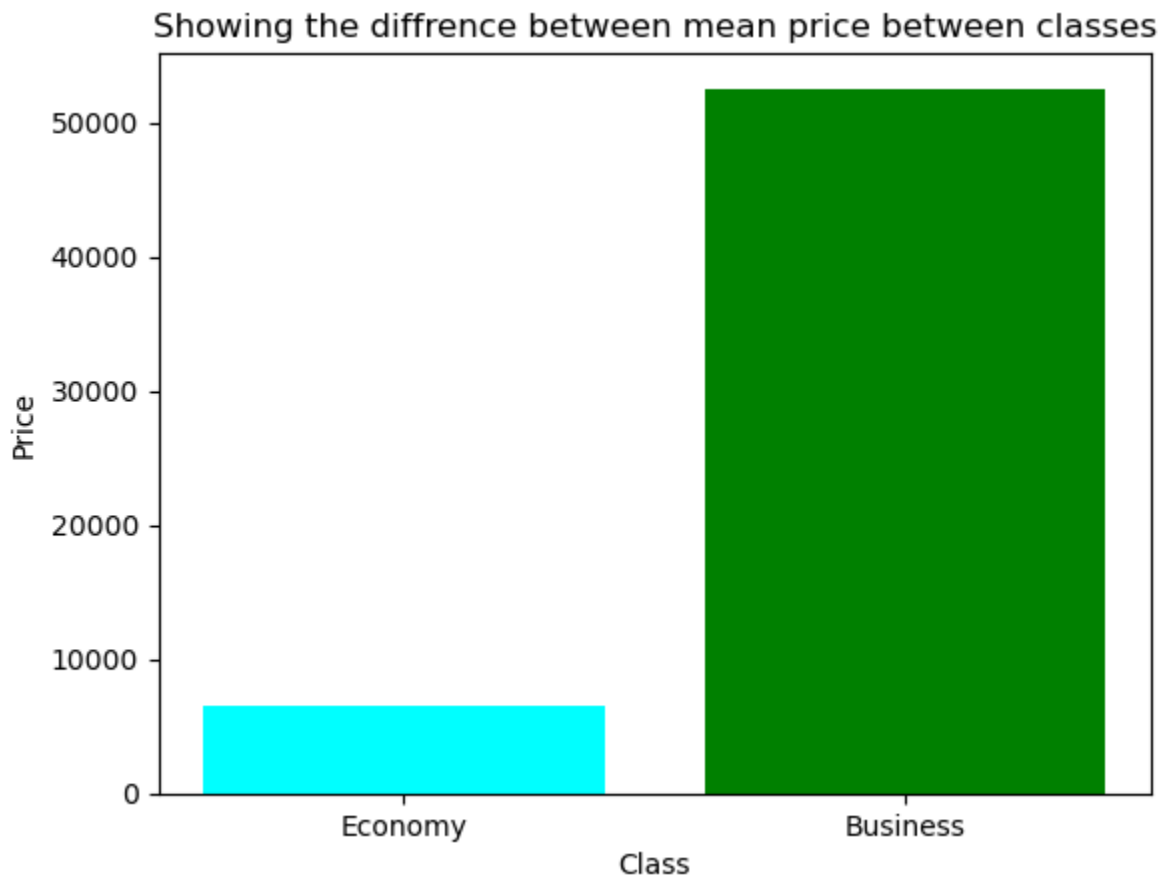


In this line chart we can see that the pricing of the tickets are higher we the flight has less than 5 days but the pricing of tickets gradually decreases when the number of days left for flight increases.

Q8. How the ticket price varies between economy and business class?

In this we have analyzed the ticket pricing based on the seat class. For this analysis we have created a bar graph

Bar graph:



In this graph we can see that average ticket price are higher for a seat in business class compared to a seat in economy class.

Conclusion:

From this project I have concluded that the ticket pricing in the aviation industry can vary a lot based on the different factors such as seat class, arrival and departure time, source and destination city, booking tickets before days left for the flight.